

Name: _____ Partner: _____ Period: _____

Two Robots, One Path

Two robots traverse the same parabolic path in a warehouse.

Robot A: $g_A(t) = (t, t^2)$, $0 \leq t \leq 3$

Robot B: $g_B(t) = (3 - t, (3 - t)^2)$, $0 \leq t \leq 3$

Robot A starts at $(0, 0)$ and Robot B starts at $(3, 9)$.

Part 1 — Tables and Shared Path

1. Complete the table for each robot.

t	Robot A position (x, y) for $g_A(t) = (t, t^2)$	Robot B position (x, y) for $g_B(t) = (3 - t, (3 - t)^2)$
0	$(0, 0)$	$(3, 9)$
1	$(1, 1)$	$(2, 4)$
2	$(2, 4)$	$(1, 1)$
3	$(3, 9)$	$(0, 0)$

2. Do both robots visit the same set of ordered pairs?

Yes. Both robots visit $\{(0, 0), (1, 1), (2, 4), (3, 9)\}$ — the same four points, just in opposite order. Robot A goes from $(0, 0)$ to $(3, 9)$; Robot B goes from $(3, 9)$ to $(0, 0)$.

Part 2 — Direction of Motion

3. At
- $t = 1$
- , is Robot A moving right or left? Up or down?

Compare $x_A(0) = 0$ and $x_A(2) = 2$: x is **increasing** $\Rightarrow$ Robot A moves **right**.Compare $y_A(0) = 0$ and $y_A(2) = 4$: y is **increasing** $\Rightarrow$ Robot A moves **up**.

4. At
- $t = 1$
- , is Robot B moving right or left? Up or down?

Compare $x_B(0) = 3$ and $x_B(2) = 1$: x is **decreasing** $\Rightarrow$ Robot B moves **left**.Compare $y_B(0) = 9$ and $y_B(2) = 1$: y is **decreasing** $\Rightarrow$ Robot B moves **down**.

5. Do the two robots move in the same direction at
- $t = 1$
- ?

No. At $t = 1$, Robot A moves right and up, while Robot B moves left and down. This illustrates EK 4.3.A.3: the same curve can be traversed in opposite directions by different parametrizations.

Teacher Note

Students sometimes assume both robots must move the same direction since they're on the same path. Q5 is the key conceptual moment — help groups articulate that direction is a property of the parametrization, not the curve itself.

Part 3 — Slope of the Curve

6. Using Robot A's data, slope between $t = 0$ and $t = 2$.

$$x_A(0) = 0 \quad x_A(2) = 2 \quad \Delta x = 2$$

$$y_A(0) = 0 \quad y_A(2) = 4 \quad \Delta y = 4$$

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{4}{2} = 2$$

7. Using Robot B's data for the same two (x, y) locations.

Robot B passes through $(0, 0)$ at $t = 3$ and through $(2, 4)$ at $t = 1$.

$$\Delta x = x_B(1) - x_B(3) = 2 - 0 = 2 \quad \Delta y = y_B(1) - y_B(3) = 4 - 0 = 4$$

$$\text{Slope} = \frac{\Delta y}{\Delta x} = 2$$

Both give slope = 2. This makes sense because slope is a property of the curve between two geometric points, not of the parametrization. Different parametrizations of the same path must give the same secant slope.

Part 4 — When Does Each Robot Arrive?

8. When does each robot pass through $(1, 1)$?

Robot A: Set $x_A(t) = t = 1 \Rightarrow t = 1$. Check: $y_A(1) = 1$ ✓

Robot B: Set $x_B(t) = 3 - t = 1 \Rightarrow t = 2$. $t = 2$. Check: $y_B(2) = 1$ ✓

9. When does each robot pass through $(0, 0)$? Through $(3, 9)$?

$(0, 0)$: Robot A at $t = 0$; Robot B at $t = 3$.

$(3, 9)$: Robot A at $t = 3$; Robot B at $t = 0$.

Teacher Note

Part 4 reinforces EK 4.3.A.2: at the same geometric point, different parametrizations arrive at different values of t . Ask groups: "Does the slope of the secant change depending on which robot you use to compute it?" (No.)